

Basics 37: Toroidal Alfvén Eigenmodes

V - Plasma waves and Alfvén physics

Toroidicity-induced gaps, coupled poloidal harmonics, discrete modes, and energetic-particle relevance

Textbook draft, version 1.0. Toroidal Alfvén eigenmodes, or TAEs, are discrete shear-Alfvén modes residing in a toroidicity-induced continuum gap. They are important in burning plasmas because fast alpha particles and beam ions can resonantly destabilize them and redistribute energetic particles. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Explain why toroidicity couples neighboring poloidal harmonics.
- Derive the approximate location of the TAE branch crossing.
- Estimate the TAE gap-center frequency.
- Describe the mixed-harmonic global eigenfunction.
- Identify energetic-particle drive and damping channels.
- Relate the reduced Module 4 benchmark to realistic TAE physics.

2 Prerequisites and reading strategy

- Basics 29, 32, and 34-36.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 From cylindrical Alfvén waves to toroidal coupling

In a straight cylinder, poloidal Fourier harmonics can be independent in an axisymmetric equilibrium. In a torus, field strength and metric coefficients vary with poloidal angle. At large aspect ratio,

$$B(\theta) \simeq B_0(1 - \epsilon \cos \theta), \quad \epsilon = \frac{r}{R_0}. \quad (3.1)$$

Multiplication by $\cos \theta$ maps $e^{im\theta}$ into $e^{i(m\pm 1)\theta}$. Toroidicity therefore couples neighboring harmonics m and $m + 1$.

4 Uncoupled continuum branches

For fixed toroidal mode number n ,

$$\omega_{A,m}(r) \simeq \frac{v_A(r)}{R_0} |m\iota(r) - n|. \quad (4.1)$$

The neighboring branch is

$$\omega_{A,m+1}(r) \simeq \frac{v_A(r)}{R_0} |(m+1)\iota(r) - n|. \quad (4.2)$$

Ignoring coupling, their magnitudes cross where the two parallel wave numbers are opposite:

$$m\iota - n = -[(m+1)\iota - n]. \quad (4.3)$$

Hence

$$\boxed{\iota_c = \frac{2n}{2m+1}}, \quad \boxed{q_c = \frac{2m+1}{2n}}. \quad (4.4)$$

5 TAE gap-center frequency

At the crossing,

$$|m\iota_c - n| = \frac{\iota_c}{2}. \quad (5.1)$$

Thus the leading gap-center estimate is

$$\omega_{\text{TAE}} \sim \frac{v_A}{2qR_0}. \quad (5.2)$$

Numerical coefficients and gap width depend on equilibrium geometry, compressibility, finite pressure, and the exact eigenvalue formulation. This estimate is a scaling law, not a precision prediction.

6 Avoided crossing and discrete global mode

Toroidal coupling produces a local two-harmonic matrix

$$\begin{bmatrix} \omega_{A,m}^2 & g_T \\ g_T & \omega_{A,m+1}^2 \end{bmatrix}. \quad (6.1)$$

The crossing opens into upper and lower continuum edges. A radial eigenmode whose frequency lies inside the gap can avoid direct intersection with the continuum over its main localization region. It has two significant poloidal components whose relative weights exchange across the coupling region.

7 Radial structure and boundary conditions

A TAE is global: its radial envelope is determined by equilibrium profiles, magnetic shear, coupling strength, and boundary conditions. Schematically,

$$-\frac{d}{dr} \left(K \frac{d\xi}{dr} \right) + H_{\text{TAE}}(r) \xi = \omega^2 M(r) \xi. \quad (7.1)$$

Gap existence alone does not guarantee a discrete mode. The radial potential and boundaries must support a normalizable solution, and coupling to other continua can still produce damping.

8 Energetic-particle resonance and drive

Fusion alpha particles or beam ions exchange energy with a TAE when their orbital phase remains coherent with the wave. A schematic resonance condition is

$$\omega - n\omega_\phi - m\omega_\theta - \ell\omega_b - \omega_d \simeq 0. \quad (8.1)$$

The energetic-particle contribution to growth depends on phase-space gradients:

$$\gamma_{\text{EP}} \sim \int d\Gamma \delta(\omega - \omega_{\text{res}}) \mathcal{W} \mathcal{D} f_{\text{EP}}. \quad (8.2)$$

The exact operator $\mathcal{D}$ and weight $\mathcal{W}$ depend on coordinates and model. A favorable gradient can drive the mode; other gradients or populations can damp it.

9 Net stability and nonlinear consequence

The linear growth rate can be organized as

$$\gamma_{\text{net}} = \gamma_{\text{EP}} - \gamma_{\text{continuum}} - \gamma_{\text{Landau}} - \gamma_{\text{coll}} - \gamma_{\text{other}}. \quad (9.1)$$

If $\gamma_{\text{net}} > 0$, amplitude grows until nonlinear effects change the distribution, mode structure, or resonance. Resulting fast-ion redistribution can flatten the driving gradient, cause prompt or resonant losses, alter alpha heating, and create localized wall loads.

10 TAE, HAE, EAE, and terminology

A TAE is specifically associated with toroidicity-induced coupling. Ellipticity-induced Alfvén eigenmodes (EAEs), noncircularity-induced modes (NAEs), and helical Alfvén eigenmodes (HAEs) arise from other geometric harmonics. In stellarators, broad three-dimensional coupling can make classification approximate. The phrase relevant to the project is *alpha-particle-driven Alfvén eigenmode*, not “alpha eigenmode.”

11 Relation to the Module 4 benchmark

The current Module 4 solver uses a reduced two-harmonic self-adjoint radial problem. It reproduces:

- uncoupled continuum branches;
- an avoided crossing and finite local gap;
- mixed-harmonic eigenfunctions;
- a discrete mode inside the gap;
- finite-volume convergence and eigenpair verification.

It does not yet include a realistic three-dimensional stellarator equilibrium, compressible continuum branches, kinetic drive, or damping. Those limitations are deliberate so the conventional reference problem remains transparent enough to validate the future eigenvalue PINN.

12 Computational laboratory

Use analytic $\iota(r)$ and density profiles to plot neighboring continuum branches, derive the crossing radius, and compare the numerical gap center with $v_A/(2qR_0)$. Solve the two-harmonic radial eigenproblem, plot harmonic weights, and scan coupling and shear. Identify parameter regions where a discrete gap mode exists.

13 Summary

- Toroidicity couples neighboring poloidal harmonics and opens a shear-Alfvén continuum gap.
- The leading TAE crossing and frequency scales follow from the parallel wave numbers.
- A TAE is a global mixed-harmonic eigenmode, not merely a local branch crossing.

- Energetic particles can drive TAEs through orbital resonances and phase-space gradients.
- The repository benchmark isolates the gap-mode mechanism before adding stellarator geometry and kinetic feedback.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
ϵ	inverse aspect ratio	r/R_0 .
ω_{TAE}	TAE frequency scale	Approximately $v_A/(2qR_0)$.
g_T	toroidal coupling	Off-diagonal harmonic coupling.
γ_{EP}	energetic-particle drive	Fast-particle contribution to growth.
$\omega_\phi, \omega_\theta, \omega_b$	orbit frequencies	Toroidal, poloidal, and bounce frequencies.
$d\Gamma$	phase-space measure	Integration measure for resonant particles.

14 Exercises

1. Derive the crossing condition $\iota_c = 2n/(2m + 1)$.
2. Derive the leading $v_A/(2qR_0)$ frequency estimate.
3. Explain why toroidicity couples m to $m \pm 1$.
4. List physical contributions to net TAE growth or damping.
5. State which TAE features the current reduced benchmark captures and which it omits.

15 References

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